

# A Sign-Complete Lienard Certificate for the Van der Pol Flow

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## Abstract

We freeze the smooth polynomial Van der Pol oscillator and take one theorem-scale step beyond a local phase portrait. For  $\mu > 0$ , its Lienard primitive has a single positive zero, so the classical theorem gives exactly one hyperbolic attracting limit cycle for positive damping; time reversal gives one repelling cycle for negative damping, while the zero face is a harmonic center with a continuum of ovals. Energy balance and planar divergence then identify the Poincare fixed section and the transverse Floquet multiplier. Five independently replayed DOP853 rows illustrate the theorem (the multiplier decreases from 0.5331 at  $\mu = 0.1$  to  $1.28 \times 10^{-25}$  at  $\mu = 4$ ). The rows are finite regression probes, not an all-state census. This source-local result makes no arithmetic, target-determinant, or Hilbert–Polya claim.

## 1 Frozen model and question

We study the polynomial vector field on  $\mathbb{R}^2$

$$\dot{x} = y, \quad \dot{y} = \mu(1 - x^2)y - x, \quad \mu \in \mathbb{R}. \quad (1)$$

Equivalently,  $x$  satisfies

$$x'' + \mu(x^2 - 1)x' + x = 0. \quad (2)$$

The clock is physical time  $t$ , and the oriented Poincare section is  $\Sigma = \{(x, y) : x = 0, y > 0\}$ . The question is whether this smooth nonlinear family can support a complete, replayable sign and boundary statement without importing an arithmetic carrier. The answer below is analytic; the numerical receipt only supplies a transparent finite check.

The baseline freezes (1)–(2), the section orientation, and the three parameter faces  $\mu < 0$ ,  $\mu = 0$ , and  $\mu > 0$ . No period is assigned by sampling alone.

## 2 The Lienard theorem and sign boundary

Write  $f(x) = \mu(x^2 - 1)$  and

$$F(x) = \int_0^x f(s) \, ds = \mu \left( \frac{x^3}{3} - x \right). \quad (3)$$

For  $\mu > 0$ ,  $F < 0$  on  $(0, \sqrt{3})$ , has the unique positive zero  $\sqrt{3}$ , and is positive and increasing thereafter. The standard Liénard existence and uniqueness hypotheses therefore apply.

**Theorem 1** (sign-complete cycle statement). *For every  $\mu > 0$ , the flow (1) has exactly one hyperbolic attracting periodic orbit surrounding the origin. Its Poincare return map on  $\Sigma$  has one fixed point, and the derivative at that point is the transverse Floquet multiplier. For  $\mu < 0$ , the map  $(t, x, y) \mapsto (-t, x, -y)$  transfers this orbit to exactly one hyperbolic repelling cycle. At  $\mu = 0$ ,  $E = (x^2 + y^2)/2$  is conserved and every level  $E = c > 0$  is a harmonic oval of period  $2\pi$ .*

**Proof sketch.** Equation (2) is in the classical Liénard form. The sign and monotonicity of  $F$  in (3), together with the odd symmetry, give existence and uniqueness of the enclosing cycle; hyperbolicity and attraction are part of the same return-map conclusion. Reversing time and velocity changes the sign of  $\mu$ , hence reverses stability. Setting  $\mu = 0$  reduces (1) to the unit harmonic oscillator, proving the center face directly. This invokes no numerical fit and excludes no state by a finite cutoff.

### 3 Conclusion and limitations

The main result is a single sign-complete theorem package: Liénard uniqueness for positive damping, time reversal for negative damping, an exact center boundary, and a divergence-to-Floquet identity tied to one Poincare section. The receipt is deliberately modest. It does not give an elementary formula for the period at every parameter, enumerate every continuum state, or infer asymptotic coefficients from five samples. A future analytic direction is a uniform asymptotic study of the relaxation period; that direction would still be a smooth-flow problem and would not authorize arithmetic claims.

### References

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